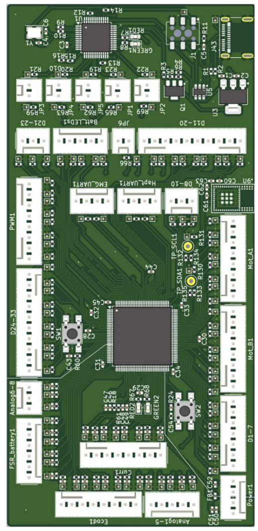
Control board : MCU Test board



- 3.3 V function (alimented through USB_C)
- Based on STM32G474QET6
    - 170 MHz
    - 128 Ko of SRAM
    - 512 Ko Flash
- Embedded debugger (ST-link V2) through USB_C
- Bluetooth module included (STM32WB1MMC)
- 2 UART connections
- 30 analog lines to ADC
- 62 digital lines
- 2 controllable LEDS
- 1 input button
- Test points on I2C Bluetooth connection
- 33 pins available for either ADC, PWM or communication protocols (I2C, SPI, USART, LPUART, Quad SPI, CAN, USB

Description:

A test board made for the needs of the arm prosthesis. Can be very versatile if more options are needed.

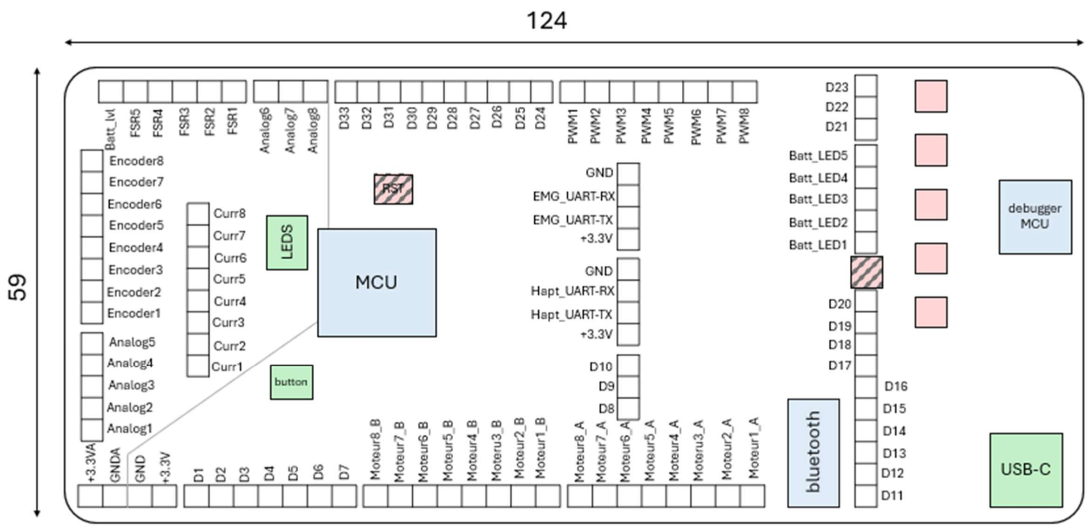
Standalone test board, with debugging and powering through USB_C port. LEDS and button for inputs and outputs.

All functions for the final board (Bluetooth, 8 motors, UART) are present.

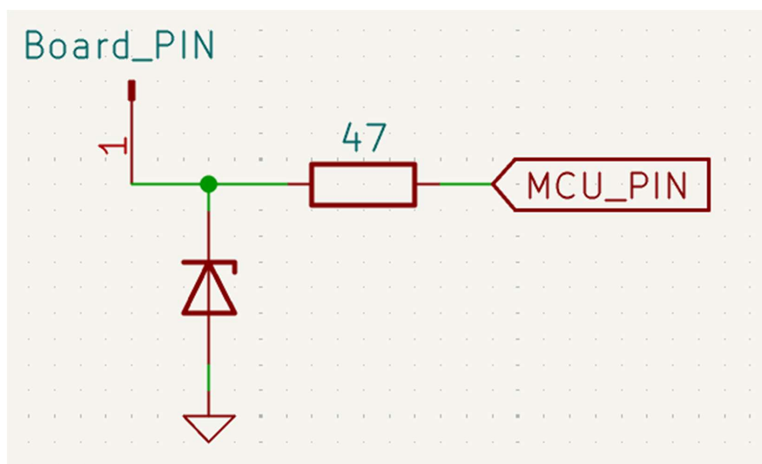
Refer to this notion page for more info on design choices : https://www.notion.so/n-pulse/Test-board-29be4f65ffef8003b585dffa8b49ce03

Index

1) Digital Pinout



a) Wiring

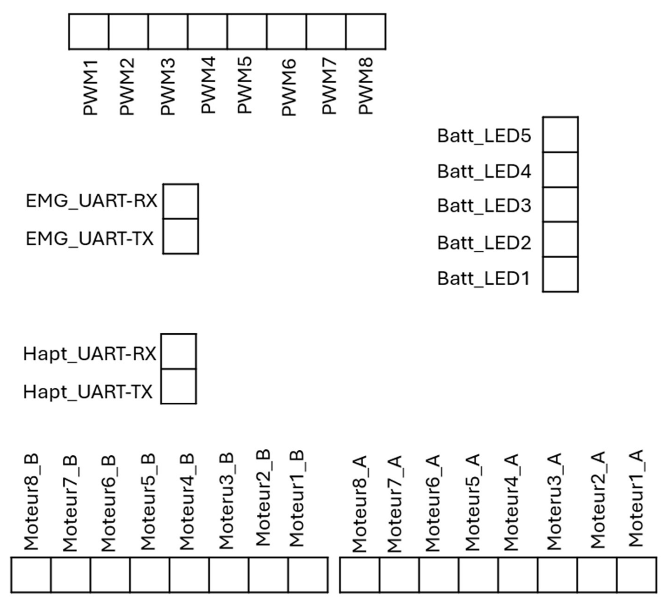


3.3V ESD protection with 3.6V breakdown voltage TVS diode and 47 Ohm resistor

b) Dedicated pins

Board PIN	MCU PIN	Function
Moteur1_A	PF5	X
Moteur2_A	PC9	X
Moteur3_A	PC8	X
Moteur4_A	PG4	X
Moteur5_A	PG3	X
Moteur6_A	PG2	X
Moteur7_A	PG1	X
Moteur8_A	PG0	X
Moteur1_B	PC7	X
Moteur2_B	PC6	X
Moteur3_B	PD15	X
Moteur4_B	PD14	X
Moteur5_B	PD13	X
Moteur6_B	PD12	X
Moteur7_B	PD11	X
Moteur8_B	PD10	X
PWM1	PE0	TIM16_CH1
PWM2	PE1	TIM20_CH4
PWM3	PE2	TIM3_CH1
PWM4	PE3	TIM3_CH2
PWM5	PE4	TIM3_CH3

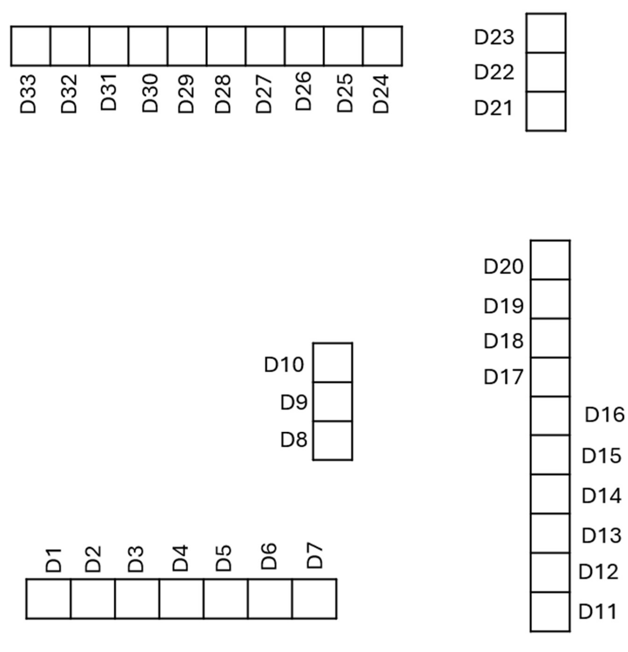
PWM6	PE5	TIM3_CH4
PWM7	PE6	TIM20_CH3N
PWM8	PC13	TIM1_CH1N
Batt_LED1	PG6	X
Batt_LED2	PD4	X
Batt_LED3	PD7	X
Batt_LED4	PB7	X
Batt_LED5	PB9	X
Hapt_UART_RX	PC11	UART4_RX
Hapt_UART_TX	PC10	UART4_TX
EMG_UART_RX	PD2	UART5_RX
EMG_UART_TX	PC12	UART5_TX



c)Other functions available

Board PIN	MCU PIN	Available function
D1	PB11	Quad_NCS/PWM/ADC
D2	PB12	CAN2_RX/ADC
D3	PB13	CAN2_TX/PWM/ADC
D4	PB14	SPI2_MISO/PWM/ADC
D5	PB15	SPI2_MOSI/PWM/ADC
D6	PD8	USART3_TX/ADC
D7	PD9	USART3_RX/ADC
D8	PA10	USART1_RX/PWM
D9	PA11	USB_DM/PWM
D10	PA12	USB_DP/PWM
D11	PF6	Quad_IO3/PWM
D12	PG5	PWM
D13	PG7	LPUART_TX
D14	PG8	LPUART_RX
D15	PG9	SPI3_SCK/PWM
D16	PD0	CAN1_RX
D17	PD1	CAN1_TX/PWM
D18	PD3	ADC/PWM
D19	PD5	USART2_TX
D20	PD6	USART3_RX/PWM
D21	PB4	SPI3_MISO/PWM
D22	PB5	SPI3_MOSI/PWM

D22	PB5	SPI3_MOSI/PWM
D23	PB6	USART1_TX/PWM
D24	PC14	Oscill32_in
D25	PC15	Oscill32_out/ADC
D26	PF3	I2C3_SCL/PWM/ADC
D27	PF4	I2C3_SDA/PWM
D28	PF7	Quad_IO2/PWM
D29	PF8	Quad_IO0/PWM
D30	PF9	Quad_IO1/PWM
D31	PF10	Quad_CLK/PWM
D32	PF0	Oscill_In/PWM/ADC
D33	PF1	SPI2_SCK/Oscill_out/ADC



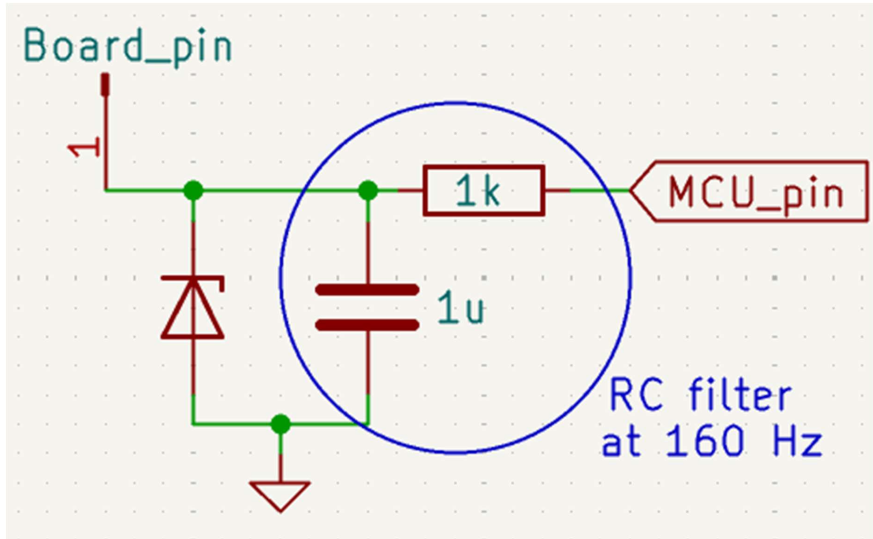
Other functions often require additional hardware (mainly pull-up / pull down resistors) that must be added out of the board

Use of an external oscillator is possible although **not** recommended. Due to the presence of 47-ohm resistors, dimensioning of capacitors must not create a filter at used frequency.

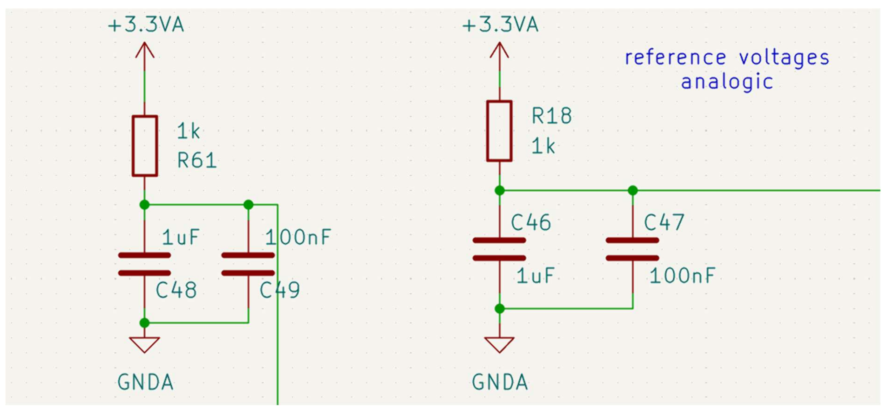
2) Analog Pinout

a) Wiring

FSR and battery level lines are filtered at 160Hz (slowly changing values) :



same ESD diode



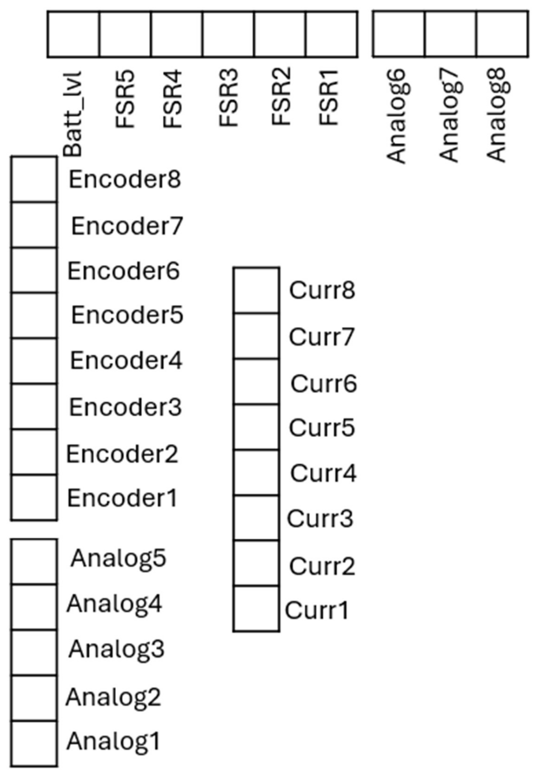
reference voltage (two separate pins to reduce impedance)

b)Pinout

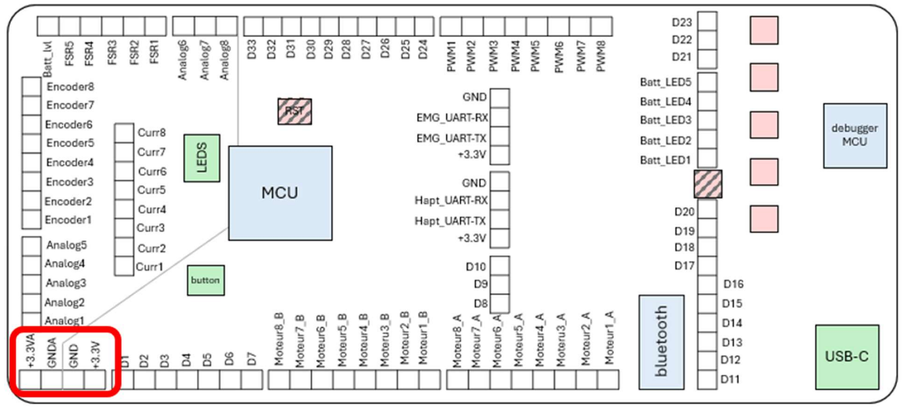
Board PIN	MCU PIN	ADC channel
Encoder1	PB1	ADC1_IN12
Encoder2	PB0	ADC1_IN15
Encoder3	PC5	ADC2_IN11
Encoder4	PC4	ADC2_IN5
Encoder5	PA7	ADC2_IN4
Encoder6	PA6	ADC2_IN3
Encoder7	PA5	ADC2_IN13
Encoder8	PA4	ADC2_IN17
Curr1	PE8	ADC3_IN6
Curr2	PE9	ADC3_IN2
Curr3	PE10	ADC3_IN14
Curr4	PE11	ADC3_IN15
Curr5	PE12	ADC3_IN16
Curr6	PE13	ADC3_IN3
Curr7	PE14	ADC4_IN1
Curr8	PE15	ADC4_IN2
Batt_lvl	PA3	ADC1_IN4
FSR1	PC2	ADC1_IN8
FSR2	PC3	ADC1_IN9
FSR3	PA0	ADC1_IN1
FSR4	PA1	ADC1_IN2
FSR5	PA2	ADC1_IN3

Analog1	PE7	ADC3_IN4
Analog2	PF14	GPIO_Analog
Analog3	PF13	GPIO_Analog
Analog4	PF11	GPIO_Analog
Analog5	PB2	ADC2_IN12
Analog6	PF2	GPIO_Analog
Analog7	PC1	ADC1_IN7
Analog8	PC0	ADC1_IN6

Analog 4 is not connected due to layout complexity



3) Power pins

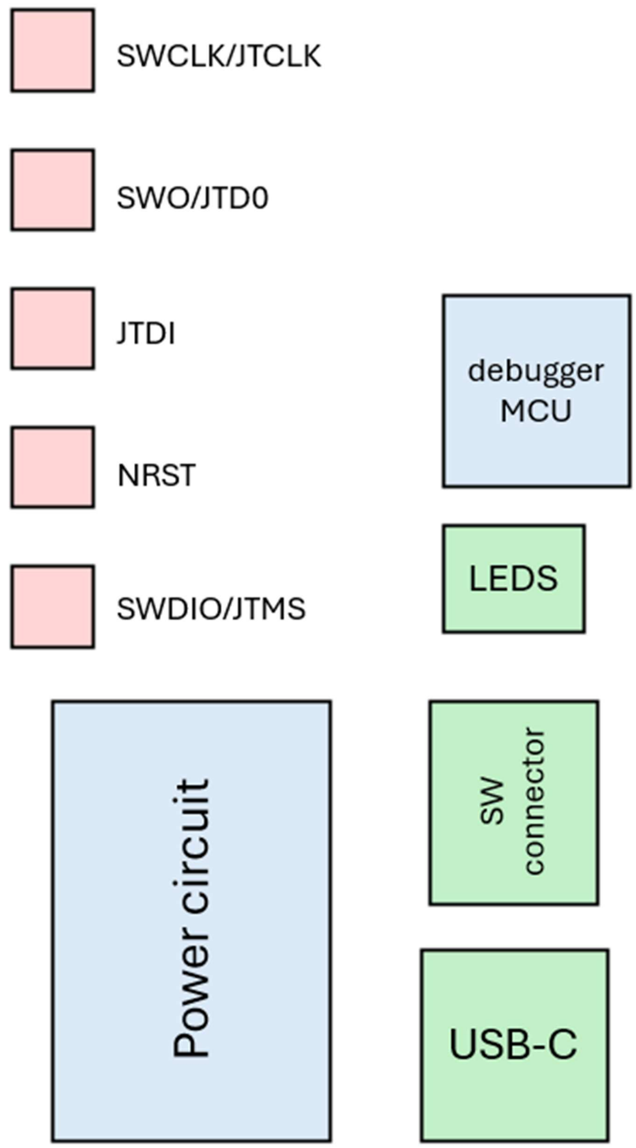


Can be used for :

- Checking the alimentation of the board
- Powering other circuits (FSRs, encoders,...)
- Alimentation of the board through GND and +3.3V pins (**not** recommended, no voltage rectifier is included, could damage components)

4) Debugger

a) General functioning



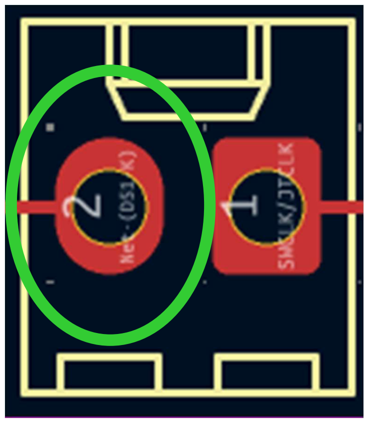
USB-C input is used as both an alimentation for the small power circuit and as a data input for the debugger

JTAG and SW protocols are both supported

Debugger must be programmed first, using the software downloaded here:

https://github.com/stlink-org/stlink/blob/testing/README.md

In case the embedded debugger does not function correctly, remove all jumpers and connect all left pins to an external ST-link



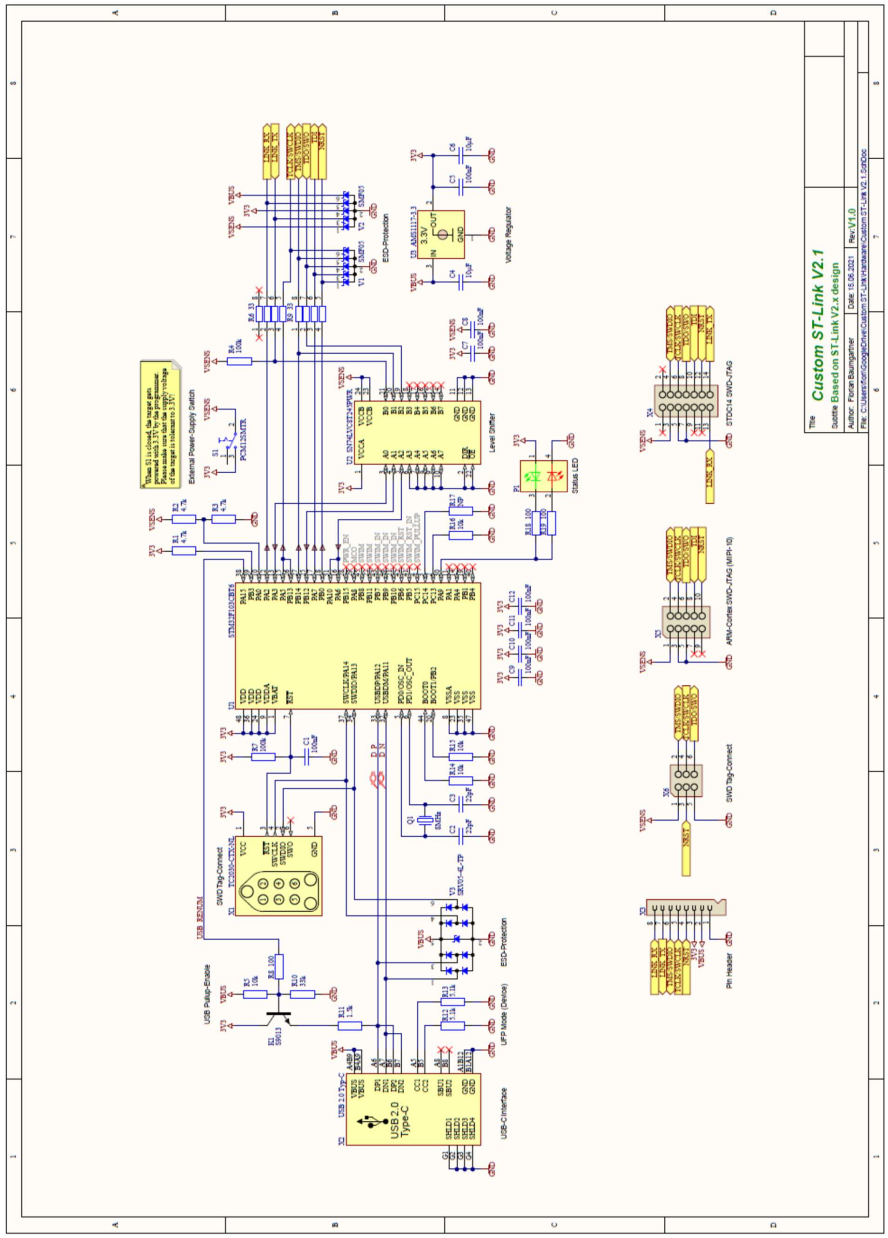
b) Source documents

Reverse engineering of ST-link v2 on this page :

https://hackaday.io/project/179054-custom-st-link-v20-v21-v30

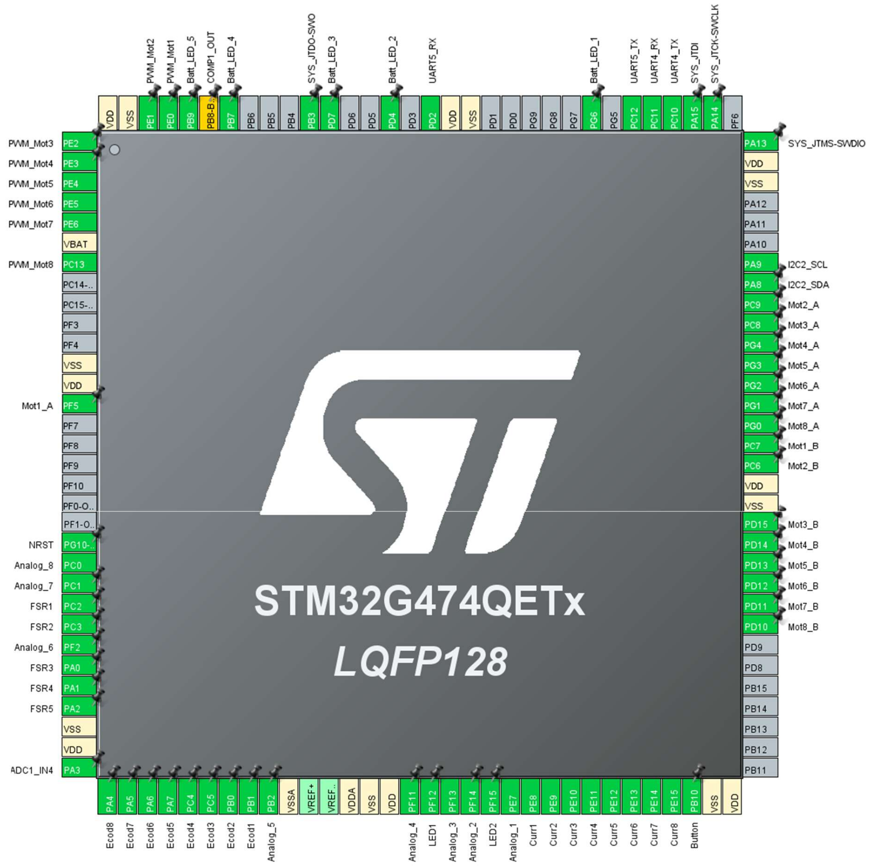
Procedure for adding the software on the debugger :

- Download STLink_Firmware.zip : https://hackaday.io/project/179054/files
- Upload files on the board using STM32CubeProgrammer and an external st-link
- Download official updated software from github : https://github.com/stlink-org/stlink/blob/testing/README.md
- Upload this official software that sould then be recognized
- Debugger is ready to use (if not, refer to the hackaday page for more information)



Custom ST-Link V2.1
Based on ST-Link V2.x design
Author: Florian Baumgartner
Date: 15.06.2021
Rev: V1.0

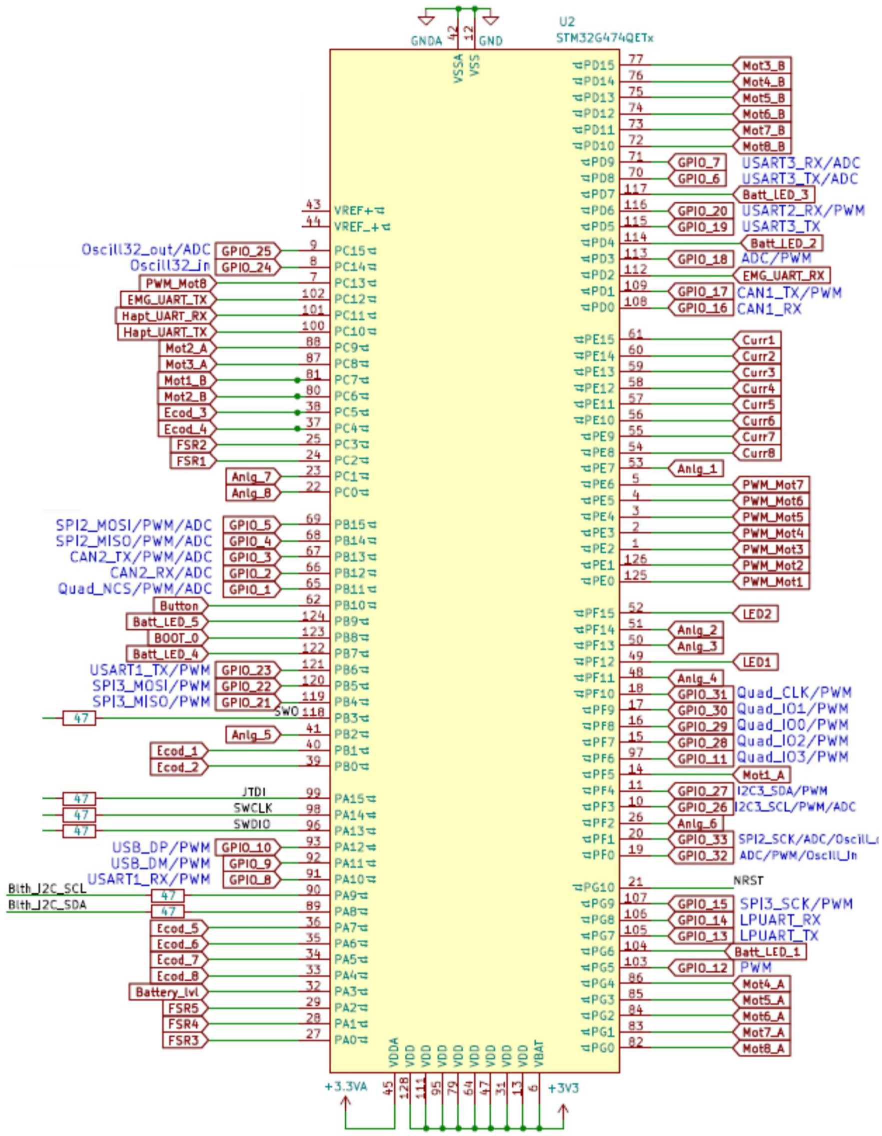
5) MCU pinout



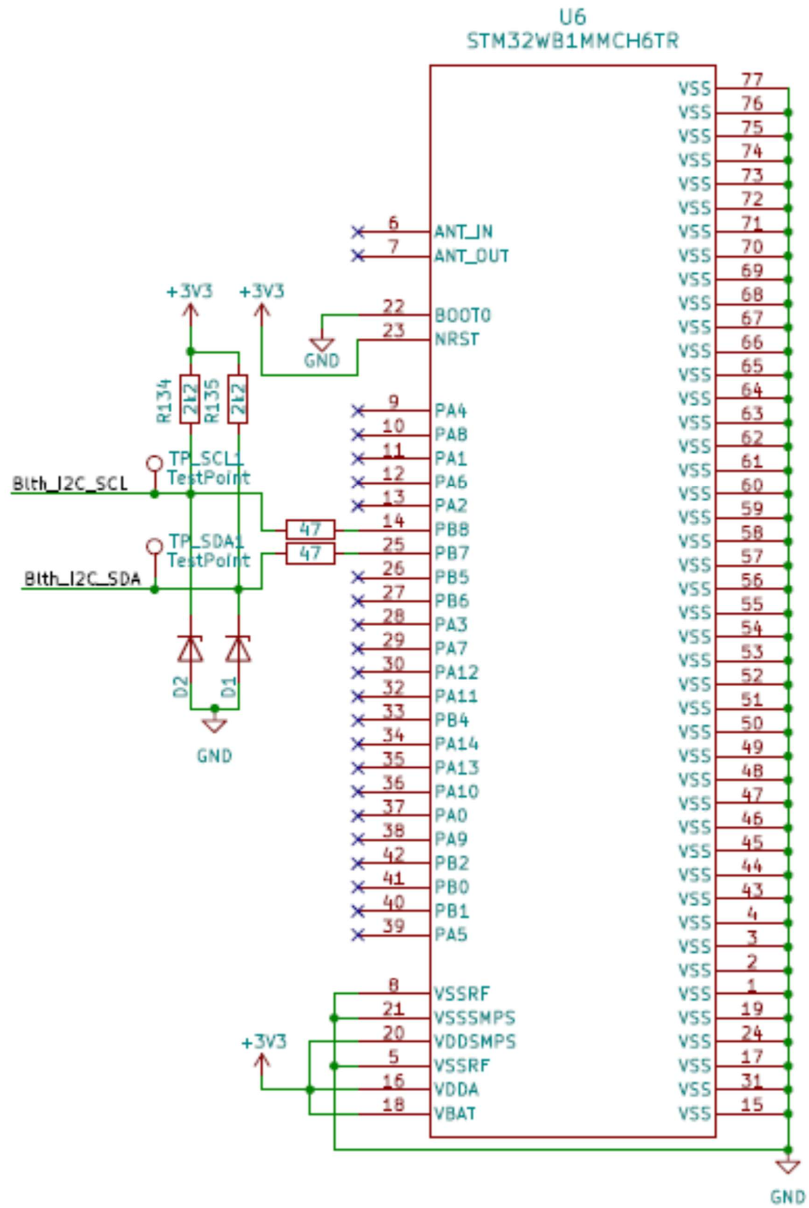
Refer to digital and analog pinouts for details

6) Schematics

a) General

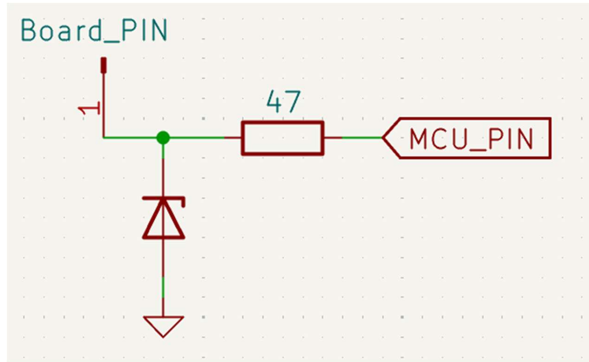


b) Bluetooth module:

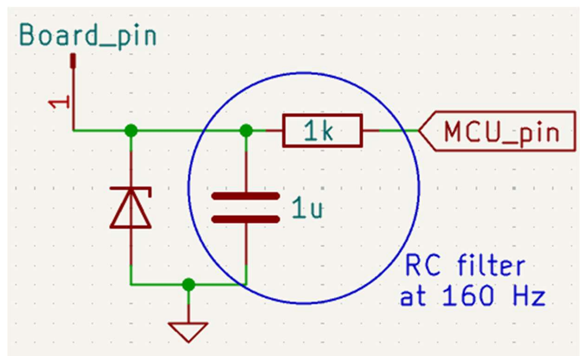


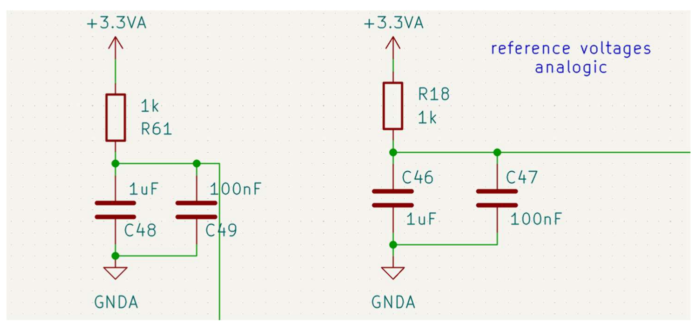
c) Analog and digital lines

Digital :

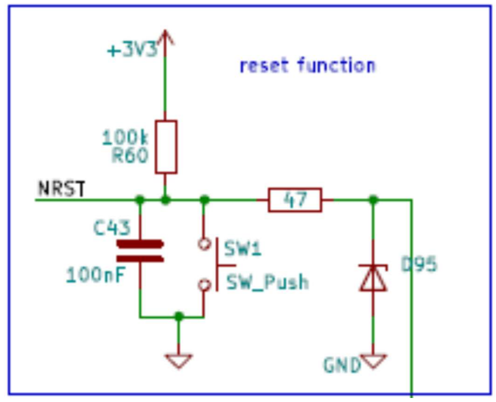


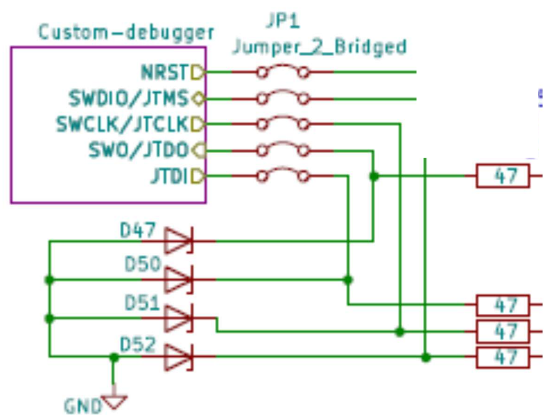
Analog:



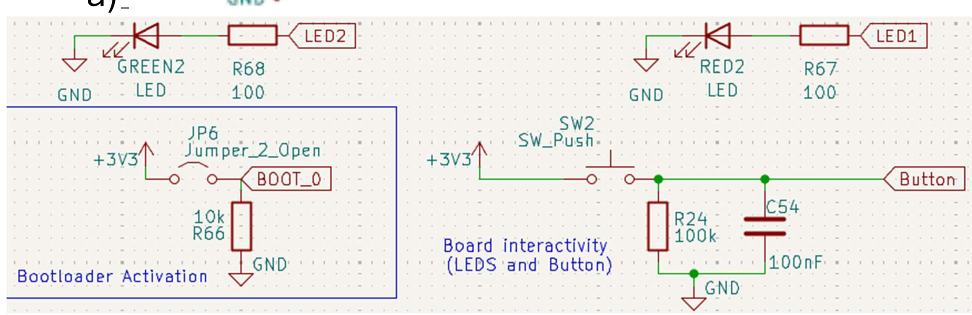


d) Connectivity

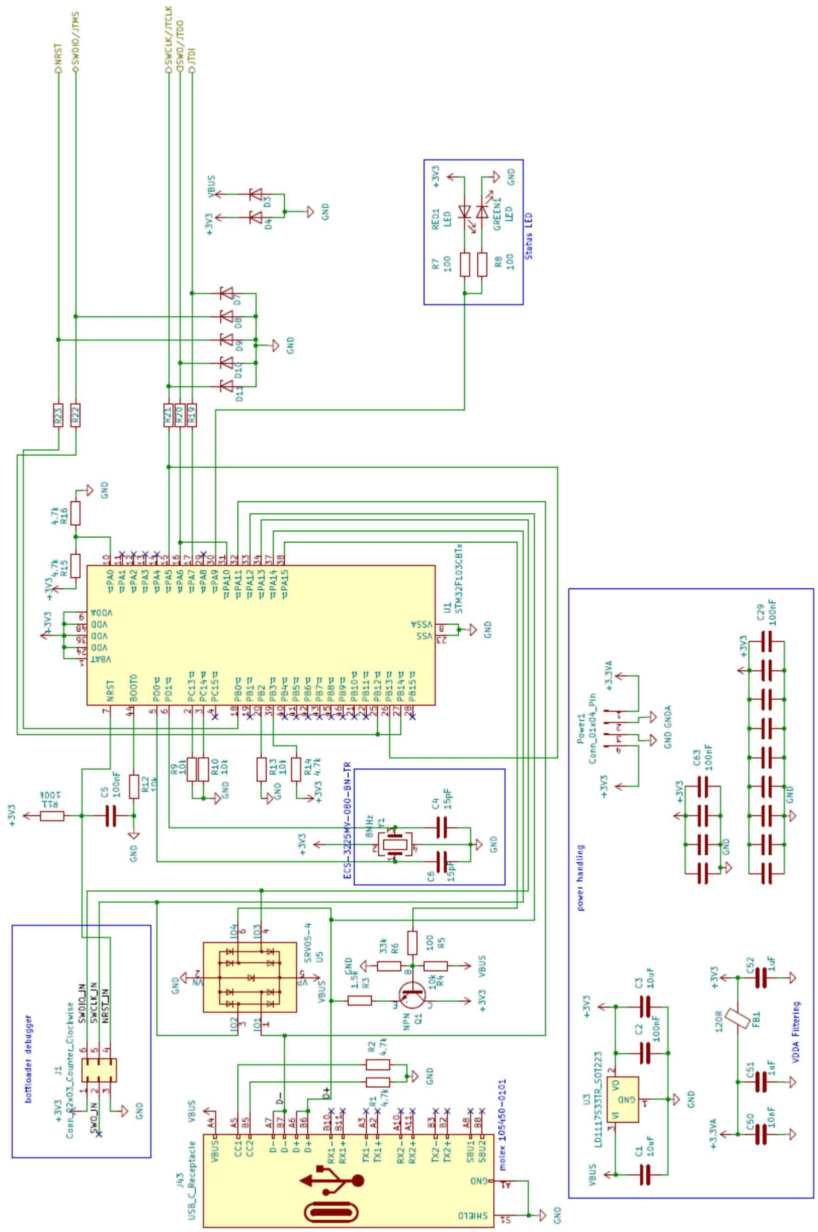




a)_

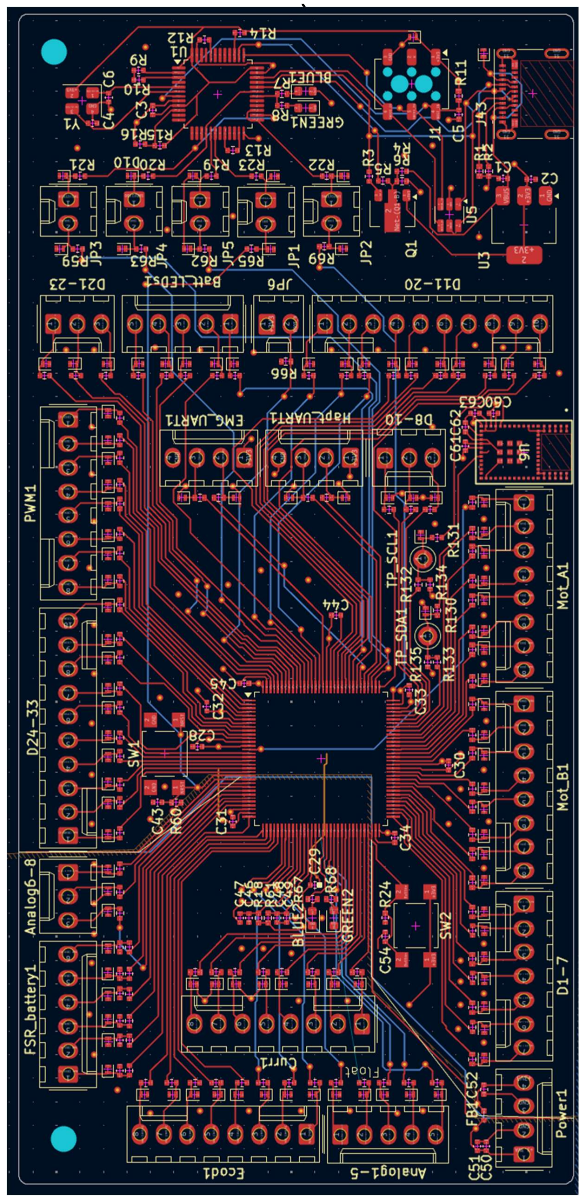


e) Debugger

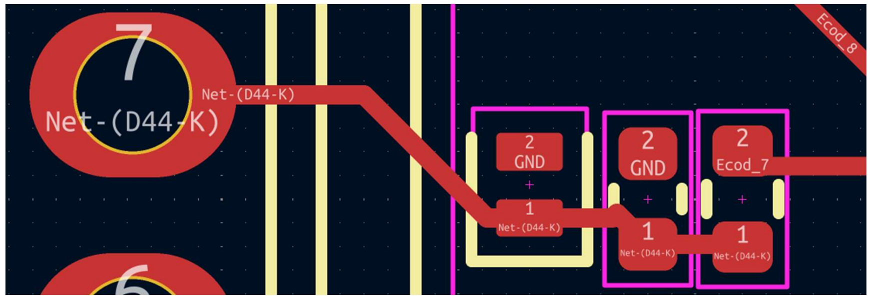

NRST
SWDIO/JTMS
SWCLK/JTCLK
SWO/JTDO
JTDI
STM32F103C8Tx
Status LED
power handling
VDDA Filtering
bottloader debugger
USB_C_Receptacle
molex 105450-0101
LD1117S33TR_SOT223
SRV05-4
ECS-3225MV-080-BN-TR

7) PCB Files

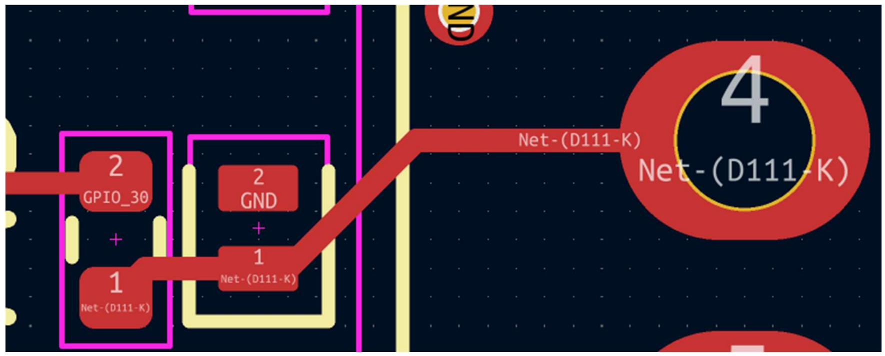
b) <u>Kicad Files</u>



Analog trace

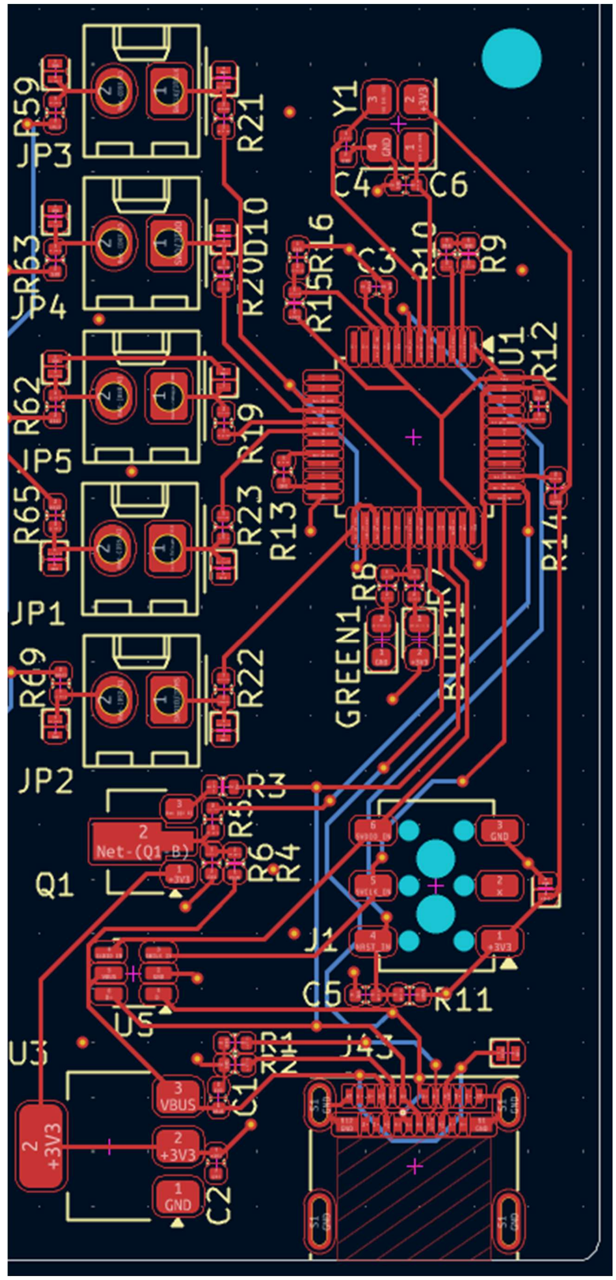


Digital trace



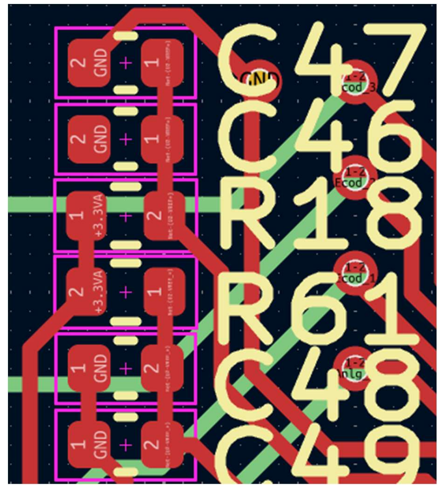
Test points

Debugger

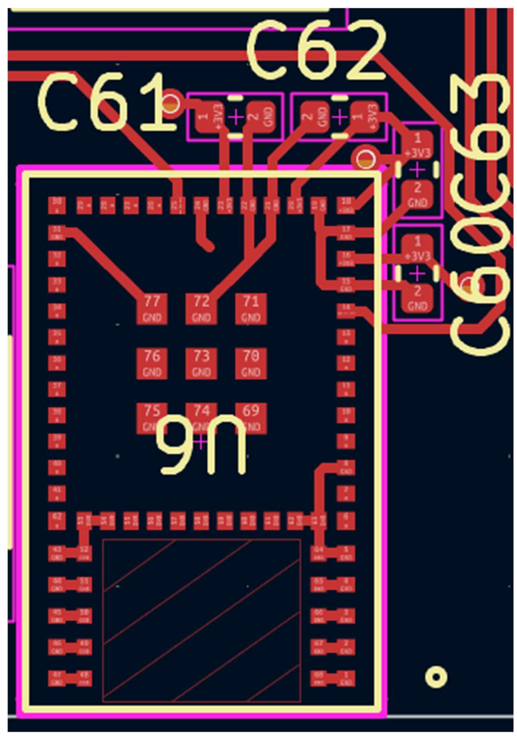


JP3
JP4
JP5
JP1
JP2
R59
R63
R62
R65
R69
R21
R20
D10
R19
R23
R13
R22
R16
R15
R10
R9
Y1
C4
C6
U1
R12
R14
R8
GREEN1
Q1
Net-(Q1-B)
R5
R6
R4
J1
U5
C5
R11
U3
VBUS
+3V3
GND
C1
C2

Voltage References

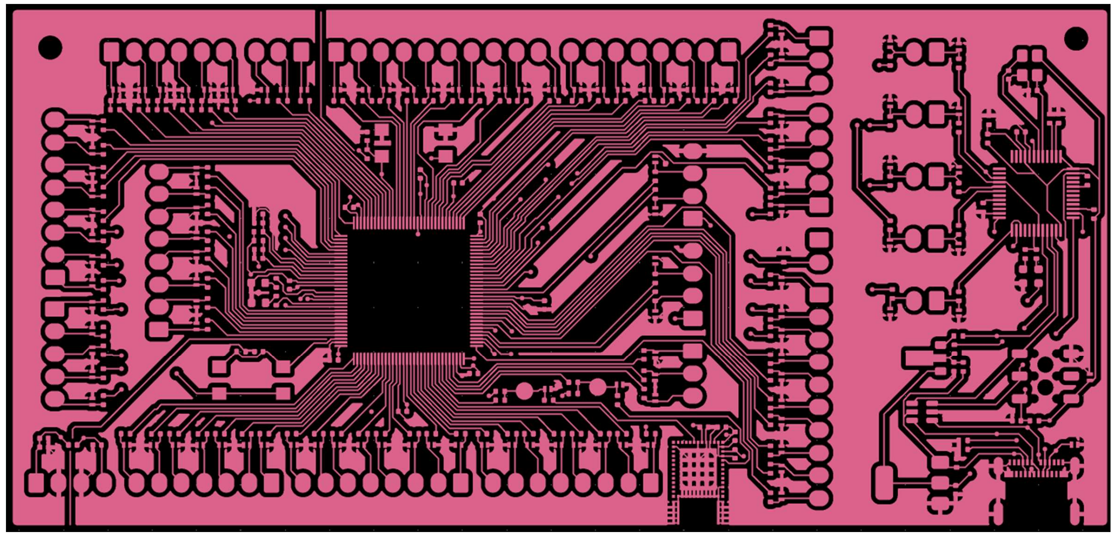


Bluetooth module

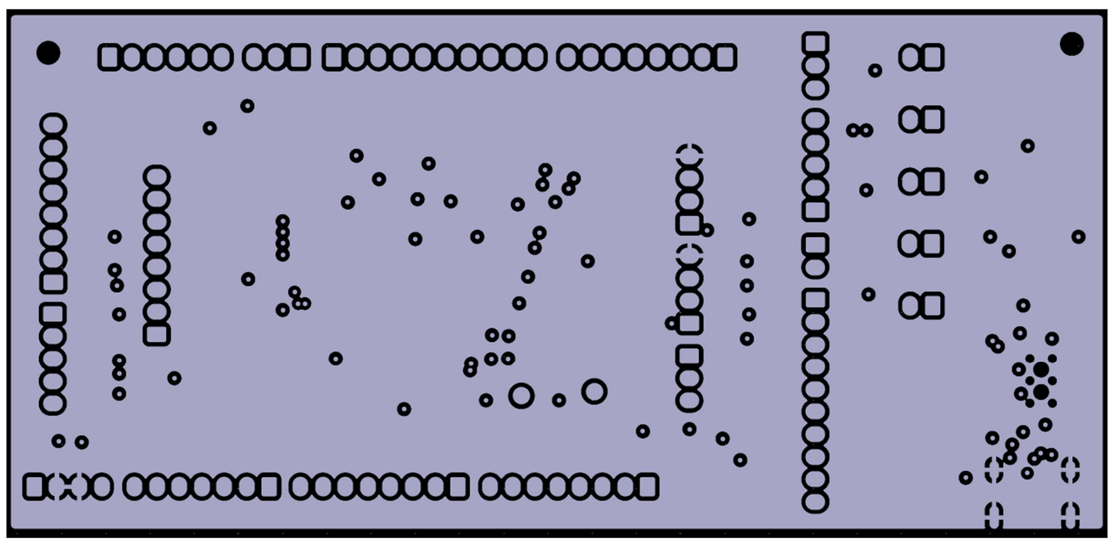


d) Gerber Files

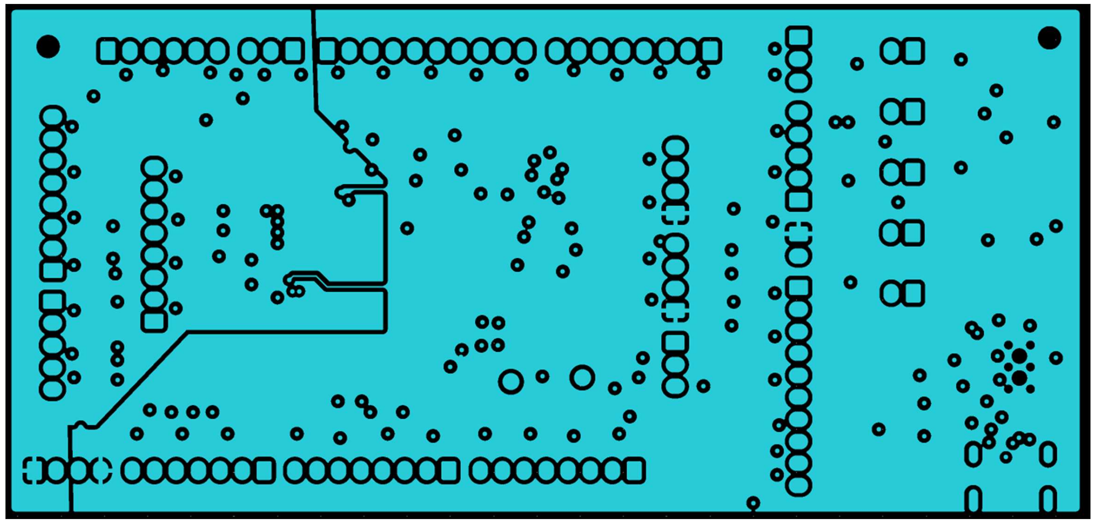
Layer 1 (data traces, GND zone)



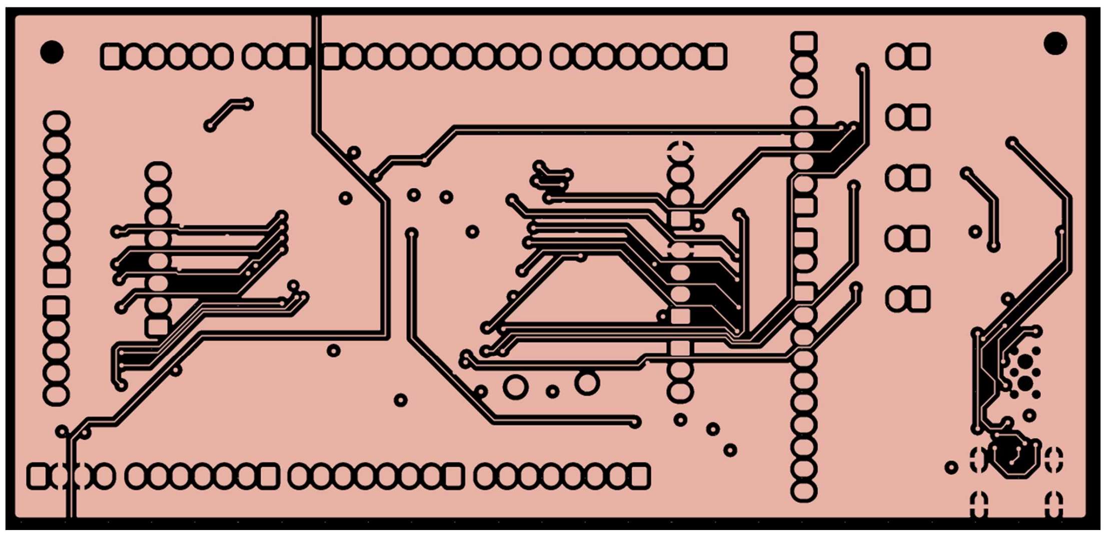
Layer 2 (GND)



Layer 3 (+3.3VA on the left, +3.3V on the right)



Layer 4 (data traces, GND zone)



8) List of components

Connectors :

USB :

- USB_C receptacle : https://www.mouser.ch/ProductDetail/Molex/105450-0101?qs=P8zB4ONU6fyZ4amolPIupQ%3D%3D

Socket :

- Harwin_M20-7810345 : https://eu.mouser.com/ProductDetail/Harwin/M20-7810345?qs=k41KVqW3ympbuqqJSo9OJw%3D%3D&srsltid=AfmBOorxiJJo-WjhTMb8x6PHU5wGUyNy78wocC7zwbb2mz9yuC2h5GuV

Pins : (kk-254 series from molex)

- 5 pins : https://www.mouser.ch/ProductDetail/Molex/22-10-2051?qs=zF2Np9wwC%252B5EI2W7nmGh%252Bw%3D%3D
- 8 pins : https://www.mouser.ch/ProductDetail/Molex/22-10-2081?qs=zF2Np9wwC%252B67nxH%252BPaWKew%3D%3D
- 4 pins : https://www.mouser.ch/ProductDetail/Molex/22-10-2041?qs=I2tGkhXPHwPlInpHG6ZNmQ%3D%3D
- 3 pins : https://www.mouser.ch/ProductDetail/Molex/22-10-2031?qs=zF2Np9wwC%252B4IPTNUsEqnkQ%3D%3D
- 7 pins : https://www.mouser.ch/ProductDetail/Molex/22-10-2071?qs=zF2Np9wwC%252B7NG4lBx4kjYA%3D%3D
- 10 pins : https://www.mouser.ch/ProductDetail/Molex/22-10-2101?qs=UAyrm%2FnZ%252BCgehm9weMkf1A%3D%3D

Jumpers :

- https://www.mouser.ch/ProductDetail/TE-Connectivity/4-530153-1?qs=kJUkXSjFC7z6%252BfUnW5DJmQ%3D%3D

Test points :

- https://www.mouser.ch/ProductDetail/Keystone-Electronics/9-5002-TL?qs=94wb96rpOOlI232itJoVLQ%3D%3D

Buttons :

- https://eu.mouser.com/ProductDetail/E-Switch/TL3305BF100QG?qs=IKkN%2F947nfDIPwqPQja%252BvQ%3D%3D

Passive components :

ESD diodes :

- DF2S5M5CT,L3F : https://www.mouser.ch/ProductDetail/Toshiba/DF2S5M5CTL3F?qs=iLbezkQI%252Bsigc8Cnmgnh7A%3D%3D
- SRV05 : https://www.mouser.ch/ProductDetail/Micro-Commercial-Components-MCC/SRV05-4L-TP?qs=SdqRYZZ9IxD6En634ibvWA%3D%3D

Resistors : (footprint 0402 in)

- 47 ohm : https://www.mouser.ch/ProductDetail/Vishay-Draloric/RCS040247R0JNED?qs=NKmfXavxMaw4EJkODaHeGQ%3D%3D
- 100 ohm : https://www.mouser.ch/ProductDetail/Vishay-Draloric/RCA0402100RFKEDHP?qs=sGAEpiMZZMtlubZbdhIBICkZD26Dv0dBylU1t1aIWbk%3D
- 1k : https://www.mouser.ch/ProductDetail/Vishay-Draloric/RCG04021K00JNED?qs=vOeJqewp7jD19ltnNDCEAA%3D%3D
- 1.5k : https://www.mouser.ch/ProductDetail/YAGEO/AC0402JR-131K5L?qs=r5DSvlrkXmJcikH0SYnX7w%3D%3D
- 2.2k : https://www.mouser.ch/ProductDetail/YAGEO/RC0402JR-132K2L?qs=fPU49sp6fCKMWFpwPhspCA%3D%3D
- 4.7k : https://www.mouser.ch/ProductDetail/YAGEO/RC0402JR-074K7L?qs=uM1JSSdPjODrMj7lsa6tLw%3D%3D
- 10k : https://www.mouser.ch/ProductDetail/YAGEO/RC0402JR-0710KL?qs=V1yeUXFNrkk4BhFptVo0Rw%3D%3D
- 33k : https://www.mouser.ch/ProductDetail/YAGEO/RC0402JR-0733KL?qs=DQ1bOpn7xPABurtcYW50LQ%3D%3D
- 100k : https://www.mouser.ch/ProductDetail/YAGEO/RC0402JR-07100KL?qs=V1yeUXFNrknA6oudwevH%252BA%3D%3D

Ferrite bead :

- BMC-Q1E0000M**:** https://www.mouser.ch/ProductDetail/TE-Connectivity-Sigma-Inductors/BMC-Q1E0000M?qs=IKkN%2F947nfDCa35q3yi9ew%3D%3D

LEDS :

- green : https://www.mouser.ch/ProductDetail/Bivar/SM0603GC?qs=RWPSkEkPOWzyFX%252BxExWW9A%3D%3D
- red : https://www.mouser.ch/ProductDetail/Bivar/SM0603RC?qs=sGAEpiMZZMvVL5Kk7ZYykbEEBd36khVH%2FDgJiJZwBC8%3D

Capacitors : (footprints 0402 in)

- 15 pF : https://www.mouser.ch/ProductDetail/Murata-Electronics/GJM1555C1HR15BB01D?qs=MYttVv2kdxlKeFWjowH6Fg%3D%3D
- 4.7nF : https://www.mouser.ch/ProductDetail/Murata-Electronics/GCG155L81H472KA01D?qs=CoAU%2FpHVZXwXCm%2FgZypGEg%3D%3D
- 10nF : https://www.mouser.ch/ProductDetail/Murata-Electronics/GCM155R71E103KA37J?qs=N3Kl9KD794TRvXRhJl8DwQ%3D%3D

- 100nF : https://www.mouser.ch/ProductDetail/Murata-Electronics/GCG155R71C104KA01D?qs=cdbOS8ANM9AFrvOKp0kMaw%3D%3D
- 1uF : https://www.mouser.ch/ProductDetail/Murata-Electronics/GRM155C81E105ME11J?qs=qSfuJ%252Bfl%2Fd7QtYUMRdX8Lw%3D%3D

Active components :

- Oscillator : ECS-3225MV : https://www.mouser.ch/ProductDetail/ECS/ECS-3225MVQ-080-BP-TR?qs=stqOd1AaK7%252BN8hLWDIRc5g%3D%3D
- Bluetooth transmitter : STM32WB1MMC https://www.st.com/en/microcontrollers-microprocessors/stm32wb1mmc.html
- Voltage regulator : https://www.mouser.ch/ProductDetail/STMicroelectronics/LD1117S33TR?qs=edoyzKMbmhntQZx4BFmoqw%3D%3D
- Transistor : https://www.mouser.ch/ProductDetail/Nexperia/BC868115?qs=me8TqzrmIYUToMzBUAqH6w%3D%3D